



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

OPC-UA PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A IoT & Edge

TOTAL: 10 QUESTIONS

Q1 What is OPC-UA and what problem in IoT & Edge does it solve? What Model

Q6 How does OPC-UA handle reliability and high availability? Reliability

Q2 What are the core components or concepts in OPC-UA? Architecture

Q7 What observability does OPC-UA provide out of the box? Lifecycle

Q3 How does OPC-UA compare to its main alternatives? Configuration

Q8 What are common performance bottlenecks in OPC-UA? Compliance

Q4 How do you get started with OPC-UA for a new project? Hardening

Q9 What security best practices apply when running OPC-UA? Testing

Q5 What configuration options most affect OPC-UA's behavior in production? Setup

Q10 What mistakes do teams commonly make when adopting OPC-UA? Tuning



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 OPC-UA is a tool or platform that

Mitigation

OPC-UA is a tool or platform that automates or simplifies a specific concern in the IoT & Edge domain, reducing manual effort and operational complexity for engineering teams.

A6 OPC-UA provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

Availability

A2 OPC-UA has key building blocks that work

Primitives

OPC-UA has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A7 OPC-UA exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

Observation

A3 OPC-UA trades off simplicity against feature richness

Hardening

OPC-UA trades off simplicity against feature richness differently than its alternatives. Choose OPC-UA when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A8 Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

Performance

A4 Install OPC-UA via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

Gotchas

A9 Run OPC-UA with the principle of least

Assessment

Run OPC-UA with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A5 Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

Configuration

A10 Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.

Documentation